

CREATING NEW CROPS

GM FOODS

As the world's population grows, scientists are seeking to make food production more efficient. One way they can do this is through changing the properties of a food-bearing plant by altering tiny sets of instructions in its cells, called genes. The resulting genetically modified (GM) crops can be made to contain more nutrients, resist pests, or need less water to grow. They are carefully tested to ensure they are safe, but many people have concerns about the long-term effects of changing these crops.



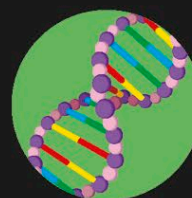
NEW FRUITS

In the 1990s, the papaya ringspot virus wiped out papaya plantations in Hawaii. A variety of GM papaya, called rainbow papaya, was developed to be resistant to the virus and help the plantations thrive once again.

Scientists are developing GM plants that can survive in space and feed astronauts.

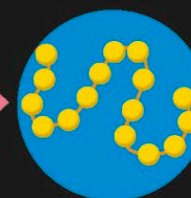
GENES

Genes are sections of DNA (see page 190), a substance found inside the cells of living things. They determine the characteristics, or traits, of living organisms. Genes are hereditary—an organism's offspring will inherit its genes from its parent or parents. In recent years, new technologies have been developed that allow scientists to modify the genes of plants and even animals.



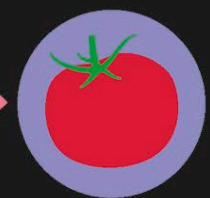
DNA

DNA is shaped like a double helix, which looks like a twisted ladder. Sections of it form genes.



Proteins

Each gene in a person's DNA is a code that tells cells to make different proteins.



Characteristic

Different proteins help make different characteristics in the organism, such as the color of a fruit.